

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)

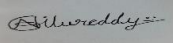
Total Marks: 50

Question Count: 5

Code of Conduct

I **G. Venkata Niranjan Reddy** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

ANSWER 1:

A Type-1 hypervisor, also called a Bare-Metal Hypervisor, runs directly on the physical hardware of a computer or server. It does not need a separate operating system to run first.

A Type-2 hypervisor, also called a Hosted Hypervisor, runs on top of an existing operating system such as Windows, Linux, or macOS.

In terms of performance, Type-1 hypervisors provide better performance because they directly manage the hardware resources without requiring an additional host operating system. Type-2 hypervisors have some performance overhead because requests pass through the host operating system.

For security, Type-1 hypervisors generally provide stronger isolation between virtual machines and have a smaller attack surface. In Type-2 virtualization, the host operating system introduces another layer that could potentially be exploited.

Regarding manageability, Type-1 hypervisors are designed for enterprise and cloud datacenters. They support features such as centralized management, live migration, high availability, and efficient resource allocation. Type-2 hypervisors are easier to install but are mainly used for desktop virtualization, testing, and development.

Factor	Type-1 Hypervisor	Type-2 Hypervisor
Performance	High performance because it runs directly on hardware	Slightly lower due to the extra host OS layer
Security	Smaller attack surface and better isolation	Host OS adds another layer that can be attacked
Manageability	Designed for centralized datacenter management	Easier for personal or testing environments
Reliability	Better suited for continuous operation	More dependent on the stability of the host OS
Examples	VMware ESXi, Microsoft Hyper-V, Xen	VirtualBox, VMware Workstation

Justification:

For mission-critical applications, I would choose a Type-1 hypervisor. It provides better performance, stronger isolation, and improved reliability. It also supports features such as live migration, high availability, and centralized resource management. Type-2 hypervisors are more suitable for development, testing, and desktop virtualization rather than large production cloud environments.

ANSWER 2:**Isolation of Finance, Healthcare, and Student Portal Workloads**

A cloud provider may have to run different applications, such as finance, healthcare, and student portals, on the same physical server. Since these applications contain different types of sensitive information, it is important to keep them separate and secure. Virtualization helps to achieve this by using a hypervisor to divide one physical server into multiple independent virtual machines (VMs).

For example, the cloud provider can create three separate VMs:

- VM 1: Finance applications and financial data
- VM 2: Healthcare applications and patient data
- VM 3: Student portal and student information

Each VM has its own virtual CPU, memory, storage, and operating system. The hypervisor controls the resources given to each VM and prevents one VM from directly accessing the resources of another VM. Therefore, even though all three workloads are running on the same physical hardware, they work as separate environments.

Virtualization can also provide isolation at different levels. Storage isolation keeps data belonging to different applications separate. Network isolation can be achieved using virtual networks, VLANs, and virtual firewalls. Access control can also be applied so that only authorized users can access sensitive resources.

Two Major Attack Surfaces

1.Hypervisor:

The hypervisor is one of the most important attack surfaces because it controls all the virtual machines. If an attacker finds a vulnerability in the hypervisor, they may try to perform a VM escape attack, where they attempt to move from one VM to the host system or access other VMs. To reduce this risk, the hypervisor should be regularly updated and securely configured.

2.Virtual Network and Management Interface:

Another major attack surface is the virtual network and management system. Poor network configuration, weak passwords, or insecure management interfaces could allow an attacker to gain unauthorized access. Proper network segmentation, firewalls, encryption, strong authentication, and role-based access control can help protect these areas.

Conclusion

In my opinion, virtualization is an effective way to allow finance, healthcare, and student workloads to share the same physical hardware while maintaining logical separation. The hypervisor provides isolation of CPU, memory, storage, and other resources, while network segmentation and security controls provide additional protection. However, the hypervisor and virtual network/management interfaces are two major attack surfaces that must be properly secured. With regular updates, strong access control, encryption, and continuous monitoring, the cloud provider can safely isolate these workloads.

ANSWER 3:

VM Host Utilization Analysis and Corrective Actions

The given VM host has CPU utilization of 92%, memory utilization of 88%, storage I/O latency of 35 ms, and an average VM boot time of 170 seconds. These values show that the host is under heavy resource pressure.

- CPU – 92%: CPU usage is very high, which may cause VMs to compete for processing resources. The likely reason is that too many or CPU-intensive VMs are running on the same host.
- Memory – 88%: High memory usage can slow down VMs and may lead to swapping when RAM becomes insufficient.
- Storage I/O latency – 35 ms: This indicates slow storage operations, possibly due to heavy disk usage or an overloaded storage system.

- VM boot time – 170 seconds: The high boot time is likely caused by the combined effect of CPU, memory, and storage bottlenecks.

Corrective Actions

1. Move some VMs to another physical host.
2. Increase CPU and RAM capacity if required.
3. Upgrade to faster SSD/NVMe storage.
4. Identify VMs consuming excessive resources.
5. Use load balancing to distribute workloads.

Conclusion

The main bottlenecks are high CPU and memory utilization, followed by storage I/O latency. Rebalancing the VMs and increasing hardware capacity can improve performance and reduce VM boot time.

ANSWER 4:

Software Defined Datacenter (SDDC) and Its Agility

A **Software Defined Datacenter (SDDC)** is a datacenter where computing, storage, and networking resources are managed and controlled mainly through software. In a traditional datacenter, most resources depend on physical hardware and manual configuration. Because of this, making changes or adding new resources can take more time.

SDDC improves agility by using virtualization and automation.

- **Compute Virtualization:** Physical servers are divided into multiple virtual machines (VMs). New VMs can be created quickly, and resources like CPU and memory can be allocated based on requirements. VMs can also be moved between servers using live migration.
- **Storage Virtualization:** Different physical storage devices can be combined into a virtual storage pool. Storage can be allocated or increased easily through software without manually changing physical devices. This improves flexibility and resource utilization.
- **Network Virtualization:** Virtual networks, switches, and firewalls can be created and configured using software. This makes it easier to change network settings, isolate workloads, and apply security policies without changing physical network equipment.

Compared with a traditional datacenter, an SDDC provides faster provisioning, easier scaling, automation, centralized management, and better flexibility. For example, when an application suddenly needs more resources, an SDDC can quickly allocate additional computing, storage, or network resources.

Conclusion

In my opinion, SDDC is more agile than a traditional datacenter because it converts physical infrastructure into flexible and programmable resources. Compute, storage, and network virtualization allow organizations to quickly respond to changing business and application requirements while reducing manual work.

ANSWER 5:

Multi-Cloud Federation and Identity Mismatch

A multi-cloud environment means using services from two or more cloud providers. In this case, the federation project failed because the cloud providers had different identity systems. The same user may have different usernames, roles, or access permissions in each cloud. Because of this mismatch, users may face login failures or may get incorrect permissions.

To solve this problem, I would use a federated identity system. A central Identity Provider (IdP) can be used to authenticate users. The cloud providers can trust this identity provider instead of maintaining separate user accounts.

The design can use federation standards such as SAML or OpenID Connect (OIDC). These allow users to log in through one trusted identity system and securely access authorized services from different cloud providers.

Single Sign-On (SSO) can also be used so that users do not need separate usernames and passwords for every cloud. The roles of users should be properly mapped between providers. For example, an administrator in the main identity system should receive only the required administrator permissions in each cloud.

For security, Multi-Factor Authentication (MFA) and the principle of least privilege should be applied. Users should only get access to the resources they actually need.

For privacy, only the minimum required user information should be shared with each cloud provider. Sensitive identity information should be encrypted during transmission, and authentication tokens should have limited lifetimes.

Conclusion

In my opinion, the main reason for the failure was the lack of a common identity system between the cloud providers. Using identity federation, SSO, SAML or OIDC, role mapping, MFA, and least-privilege access can solve the identity problem. At the same time, minimum data sharing and encryption can protect user privacy. This design provides a secure and convenient way to manage users across multiple cloud platforms.